

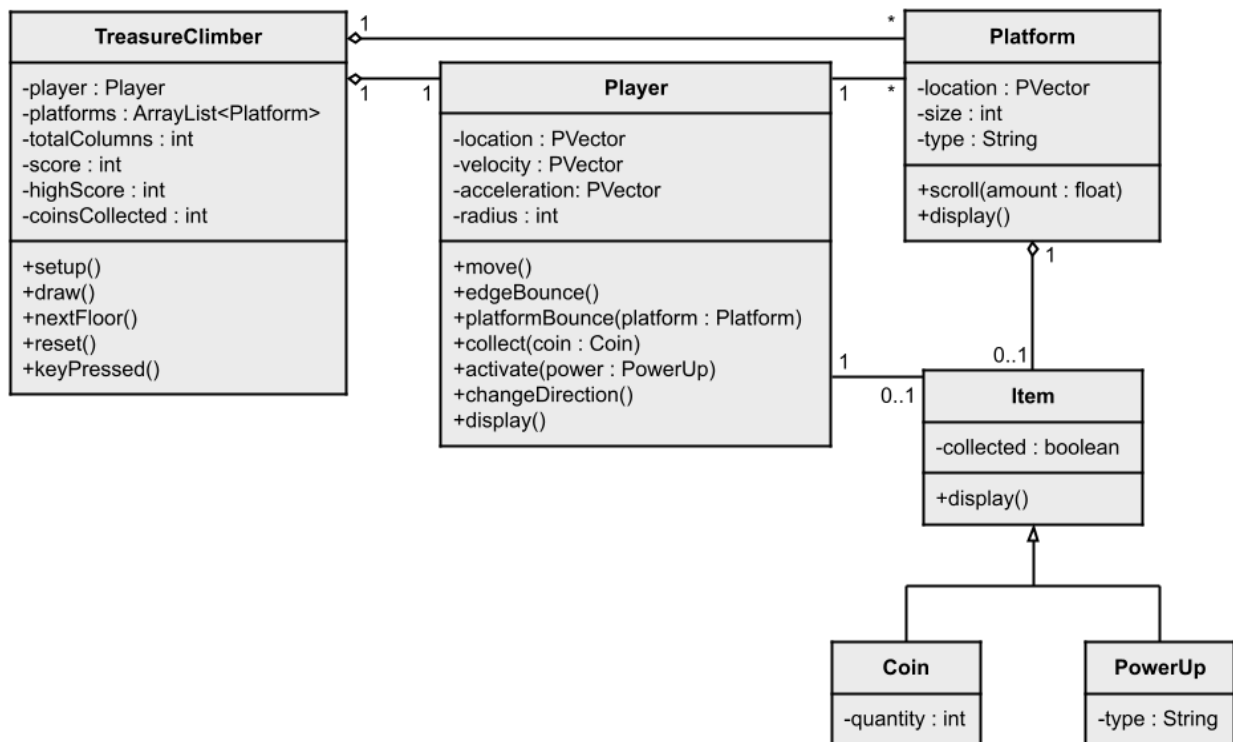
Intro

Period: 2
Name: Ivan Zheng
Group: Mine Director
Project Title: Treasure Climber

Description

Treasure Climber is a game inspired by Doodle Jump. The player controls a character by having it move left and right to help it land on platforms to climb higher and higher. A final score will be calculated based on how high the player goes and how many coins the player collects. The game ends if the player falls below the bottom of the screen.

UML Diagram



Directions

The player spawns at the bottom of the mine and the character does not start until the player presses spacebar. Once the player starts, the character will keep moving in that direction until it hits a side of the screen or the player presses the spacebar again. As the player climbs higher, the character moves faster and faster, making the game progressively more difficult.

The objective of the game is to get the highest score possible. This can be achieved by climbing higher and collecting more coins. As of now, the game is planned to be endless.

Functionalities

- character with physics
 - left and right direction is swapped by pressing spacebar
 - keyPressed() in TreasureClimber, changeDirection() in Player
 - bounces off edges of the screen
 - bounces off platforms using Axis-Aligned Bounding Box (AABB) collision detection
 - starts immediately
- starting platforms of slightly different sizes arranged in seven columns
 - rectMode(CENTER) is used for locations
 - added totalColumns variable in TreasureClimber

Planned Functionalities (in order)

- player doesn't move until user presses spacebar (starts moving right)
- vertical scrolling (platforms move down when player moves up)
- coins on platforms to collect
- scoring system
- powerups on platforms
 - timer: reduces speed
 - jetpack: propels player upwards
 - magnet: attracts coins
 - shield: grants extra life
- different platform types
 - trapdoor: disappears when landed on
 - spikes: ends game when landed on
 - conveyor: moves back and forth